

technology has been used to enable wireless control. The device is built around an Arduino Uno (MOD1), Bluetooth module HC-05 (MOD2), L298N motor driver (IC1), two 12V, 500 RPM geared DC motors (M1, M2), 100cm diameter wheels with 6mm shafts, a breadboard, jumper wires, USB cable, wooden plank, battery holder, and four 18650 Li-ion

battery cells (3.7V, 2000mAh each).

The L298N-based motor driver module (2A) is a high-power driver suitable for controlling both DC and stepper motors. It can drive up to four DC motors or two DC motors with directional and speed control. The Arduino Uno features digital pins D2 through D13 and analogue pins A0 through A7.

```

1  char junk;
2  String inputString="";
3  void setup()                // run once, when the sketch starts
4  {
5      Serial.begin(9600);      // set the baud rate to 9600, same should be
6      pinMode(13, OUTPUT);
7      pinMode(12, OUTPUT);
8      pinMode(11, OUTPUT);
9      pinMode(7, OUTPUT);
10     pinMode(8, OUTPUT);
11     pinMode(9, OUTPUT);
12     pinMode(10, OUTPUT);
13 }
14 void loop()
15 {
16     if(Serial.available()){
17         while(Serial.available()){
18             {
19                 char inChar = (char)Serial.read(); //read the input
20                 inputString += inChar;           //make a string of the characters coming
21             }
22             Serial.println(inputString);
23             while (Serial.available() > 0)
24             { junk = Serial.read(); } // clear the serial buffer
25             if(inputString == "F"){ //in case of 'a' turn the LED on
26                 digitalWrite(13, HIGH);
27                 digitalWrite(12, LOW);
28                 digitalWrite(11, HIGH);
29                 analogWrite(9, 255);
30                 digitalWrite(7, HIGH);
31                 digitalWrite(8, LOW);
32                 analogWrite(10, 255);
33             }
34             else if(inputString == "B"){ //in case of 'b' turn the LED off
35                 digitalWrite(13, HIGH);
36                 digitalWrite(12, HIGH);
37                 digitalWrite(11, LOW);
38                 analogWrite(9, 255);
39                 digitalWrite(7, LOW);

```

Fig. 3: Code snippet of command set used to control the robot

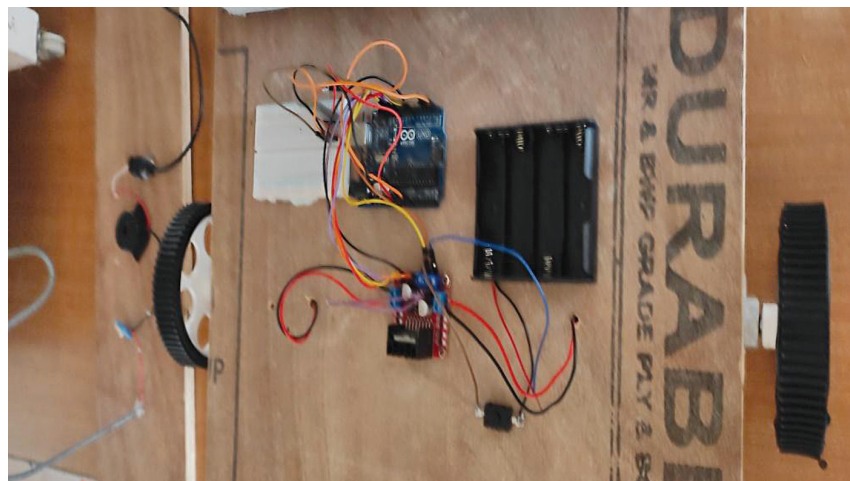


Fig. 4: Assembly of motor driver and Arduino mounted on the chassis

BILL OF MATERIALS

Components	Quantity
Arduino Uno (MOD1)	1
Bluetooth HC-05 (MOD2)	1
L298 motor driver (IC1)	1
12V, 500 RPM geared DC motor (M1, M2)	2
Battery holder	1
USB cable	1
Breadboard	1
Battery 18650 3.7V 2000mAh	4
Wooden plank	1
100mm diameter tyres and robot wheels with 6mm shaft	2 set
Small castor wheel	1

The HC-05 Bluetooth module connects to the Arduino Uno. The Tx and Rx pins of the HC-05 are connected to pins 0 and 1 (Rx and Tx) of the Arduino, respectively. Vcc is connected to the 5V pin, and GND to the GND pin.

Pins D9 and D10 of the Arduino Uno connect to Enable A and Enable B of the L298N driver. Pins D8 and D7 are connected to IN1 and IN2, while pins D11 and D12 are connected to

IN4 and IN3 of the driver. The DC motors are connected to the motor output pins of the driver module.

Software

Programming of the robot is carried out using the Arduino IDE. The Arduino Uno sends control signals to the motor driver, which in turn drives the motors to move the trolley forward, backward, left, or right. Commands are received from an Android mobile application via the HC-05 Bluetooth module.

The Android application can be downloaded from the Google Play Store. When commands are sent via the app:

- 'F' or 'f' moves the trolley forward
- 'B' or 'b' moves it backward
- 'L' or 'l' turns it left
- 'R' or 'r' turns it right
- 'S' or 's' stops the trolley

To begin operation, upload the code to the Arduino Uno by selecting the correct board and COM port in the IDE. Ensure Arduino Uno is selected as the board. Fig. 3 shows the code snippet of command set used to control the robot.

Construction and testing

Begin by uploading the source code to the Arduino Uno. Next, connect